

| Safety-alignment<br>Dataset Size (→) |         | None                                   | Full (7.7K) | 5K       |        |               | 1K       |        |               |
|--------------------------------------|---------|----------------------------------------|-------------|----------|--------|---------------|----------|--------|---------------|
|                                      |         |                                        |             | High-Sim | Random | Low-Sim       | High-Sim | Random | Low-Sim       |
| Initial                              | Utility | 7.48                                   | 7.59        | 7.68     | 7.34   | 7.76          | 7.66     | 7.41   | 7.74          |
|                                      | HS      | 71.00%                                 | 9.00%       | 16.67%   | 11.33% | 10.33%        | 30.00%   | 28.67% | 24.67%        |
| Dataset                              |         | Downstream Fine-tuning (Harmful Tasks) |             |          |        |               |          |        |               |
| List                                 | HS      | 77.33%                                 | 67.67%      | 70.33%   | 69.67% | <b>67.33%</b> | 78.67%   | 73.67% | <b>71.00%</b> |
| Pure Bad                             | HS      | 82.33%                                 | 73.33%      | 80.67%   | 78.33% | <b>76.33%</b> | 89.33%   | 84.00% | <b>77.67%</b> |
| Dataset                              |         | Downstream Fine-tuning (Benign Tasks)  |             |          |        |               |          |        |               |
| Alpaca                               | Utility | 5.75                                   | 6.36        | 5.68     | 6.34   | 5.96          | 5.74     | 6.33   | 5.88          |
|                                      | HS      | 49.67%                                 | 38.00%      | 52.84%   | 53.33% | <b>48.67%</b> | 56.00%   | 59.33% | <b>50.33%</b> |
| SAMSum                               | Utility | 50.74%                                 | 52.26%      | 54.53%   | 52.79% | 52.22%        | 56.54%   | 58.51% | 54.66%        |
|                                      | HS      | 85.00%                                 | 53.33%      | 80.33%   | 76.33% | <b>70.00%</b> | 85.67%   | 80.00% | <b>77.00%</b> |

Note. For High-Sim’s and Low-Sim’s Initial models, we report the average score across four target downstream datasets

**Table 2: Utility/harmfulness before/after downstream fine-tuning of LLAMA-2-13B**

downstream fine-tuning tasks. It is also worth noting that models tended to be safer, as indicated by lower HS, when a larger safety-alignment dataset was used.

**Upstream Plus Downstream Defenses Strengthen Guardrails More Than Either Alone.** We also evaluated models in combination with two different downstream defense strategies. Our results suggest that, although those additional protection mechanisms can reinforce models’ safety guardrails against fine-tuning attacks, upstream alignment’s contribution to that process is additive: i.e., Low-Sim yielded better safety than High-Sim, irrespective of which downstream defense was in play.

## 5 Discussion

**Implications.** Our findings underscore the critical role of dataset privacy and representation similarity in establishing robust safety guardrails for LLMs. We have shown that high representational similarity between upstream alignment data and downstream fine-tuning tasks can markedly compromise safety guardrails, even when the fine-tuning data is entirely benign. As summarized in Figure 4a, increased similarity is correlated with increased vulnerability to jailbreaks, while lower similarity enhances the robustness of safety constraints.

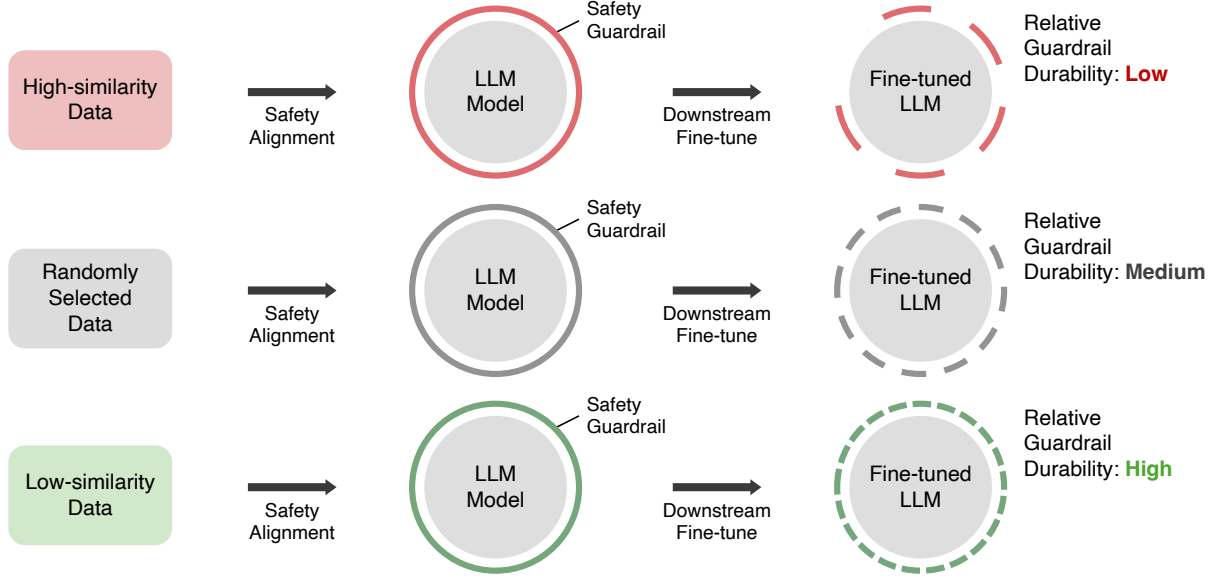
This has profound implications for the responsible development and regulation of LLMs. In particular, it suggests that privacy-preserving alignment processes are not merely a matter of ethical

data governance, but are also directly linked to the structural integrity of safety mechanisms. Public release or careless handling of alignment datasets could enable adversaries to construct fine-tuning tasks that deliberately mimic original data distributions, thereby dismantling models’ guardrails post-alignment. Our results extend emerging discussions around regulatory accountability and safety disclosures for foundation models (Kshetri, 2024).

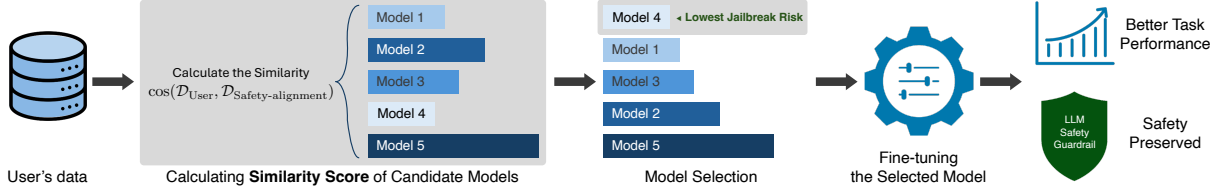
**Novel Insights.** This study also advances the new perspective that *representation similarity is a quantifiable and actionable risk factor for models’ jailbreak vulnerability*. Prior work has predominantly focused on architectural defenses or adversarial training. By contrast, our approach suggests that LLM robustness can be enhanced preemptively through informed dataset-engineering and model-selection strategies.

In practice, fine-tuning service providers like OpenAI and Anthropic can leverage our findings by computing representation similarity between upstream alignment corpora and candidate downstream datasets. Models that are too aligned (or misaligned) with user-provided data in representation space can be flagged. We illustrate this approach in Figure 4b, outlining a simple pipeline that enables providers to make safer deployment decisions—either by rejecting unsafe fine-tuning requests or routing them to models aligned with more orthogonal data distributions.

Finally, our method is complementary to existing safety defenses. For example, similarity-aware model selection can be used in conjunction with



(a) Unsurprisingly, given that they all had low harm scores before downstream fine-tuning, the three subsets produced equally safe guardrails after safety alignment. However, those guardrails’ durability varied with different task similarities: i.e., High-Sim weakened guardrails (red) most severely; Random resulted in medium durability (gray); and Low-Sim preserved more safety (green)



(b) Given a user-provided dataset, providers compute representation similarity across a pool of safety-aligned candidate models. Models with low similarity to the downstream task data are flagged as lower risk for safety degradation. The selected model is then fine-tuned, resulting in improved task performance while preserving safety guardrails and reducing harmful outputs. This approach enables fine-tuning service providers to proactively mitigate jailbreak vulnerabilities through informed model selection

**Figure 4: (a) Impact of safety-alignment data similarity on LLM guardrail durability; (b) Similarity-aware model selection pipeline for safer fine-tuning**

post-hoc pruning (Huang et al., 2025a), constraint-based fine-tuning (Hsu et al., 2024), or residual output filters (Ji et al., 2024a), forming a layered strategy that strengthens robustness throughout the full deployment pipeline.

**Future Directions.** This work opens several paths for further exploration. First, our basic approach of studying safety guardrails from their formation could be combined with task vector analysis to pinpoint the internal representations and neurons most susceptible to erosion during fine-tuning (Ilharco et al., 2023; Liu et al., 2025c). Analyzing differences in those vectors between High-Sim and Low-Sim conditions would likely provide important insights into the neural underpinnings of durable safety.

Second, although we focused here on safety guardrails targeting harmful outputs, our methodology can be extended to study other forms of align-

ment guardrails across domains including factuality, fairness, and helpfulness (Rebedea et al., 2023; Kang and Li, 2025; GuardrailsAI, 2024).

Finally, given that multimodal and reasoning-intensive models become increasingly prevalent, their safety remains a critical issue (Huang et al., 2025d; Wang et al., 2025; Zhou et al., 2025; Fang et al., 2025; Jiang et al., 2025). Future work could usefully examine how alignment similarity manifests in more complex modalities—such as long-form reasoning, image-text pairs, or video-language inputs—where representational entanglement may introduce new vulnerabilities.

## 6 Conclusion

This work has identified representation similarity between upstream alignment data and downstream fine-tuning tasks as a critical yet previously overlooked factor in the erosion of LLMs’ safety guardrails. Our experiments demonstrated

that high-similarity datasets substantially increase a model’s susceptibility to jailbreaks, even when downstream data is entirely benign. Conversely, dissimilarity fosters safety over and above the positive impact of existing downstream defense systems. These findings carry broad implications for LLM development and deployment, and our analysis offers a practical framework for safe model selection during fine-tuning and proactive alignment management. As LLMs become increasingly embedded in critical decision-making systems, durable safety must move beyond reactive patching and toward alignment-aware training and deployment. This study has charted a course for this transition toward more robust, trustworthy, and secure language models.

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